

Inverse Design of Light-Trapping Layers

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Abstract

This report details the surrogate optimization and active learning approaches used to inversely design light-trapping layers.

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1 Introduction

Introduce the problem of light-trapping layers here [1].

2 Methods

2.1 RCWA and Classical Solvers

2.2 Machine Learning Surrogates

2.3 Optimization Strategies

3 Results

Present the performance of the surrogate models and the final inverse design outcomes.

Figure 1: Placeholder for a results figure.

4 Conclusion

Summarize the findings and future work.

References

- [1] A. Author and B. Writer. A study on light-trapping nanostructures. *Journal of Nanophotonics*, 10:1–10, 2026.